

Nonlinear predictive control of a boiler-turbine unit: A state-space approach with successive on-line model linearization and quadratic optimization

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Abstract: This paper details development of a Model Predictive Control (MPC) algorithm for a boiler-turbine unit, which is a nonlinear multiple-input multiple-output process. The control objective is to follow set-point changes imposed on two state (output) variables and to satisfy constraints imposed on three inputs and one output. In order to obtain a computationally efficient control scheme, the state-space model is successively relinearized on-line for the current operating point and used for prediction. In consequence, the future control policy is easily calculated from a quadratic optimization problem. For state estimation the extended Kalman filter is used. It is demonstrated that the MPC strategy based on constant linear models does not work satisfactorily for the boiler-turbine unit whereas the discussed algorithm with on-line successive model linearisation gives practically the same trajectories as the truly nonlinear MPC controller with nonlinear optimization repeated at each sampling instant.

Keywords: Process control; Model predictive control; Boiler-turbine unit; On-line successive linearization

Table of Contents

Modeling.....	2
Section 1: Equilibrium verification and operating point analysis.....	3
Section 2: Continuous-time step response analysis.....	6
Input-output coupling structure.....	6
Individual input-output dynamics.....	6

Fuel valve 's effect on outputs.....	6
Steam valve 's effect on outputs.....	7
Feed-water valve 's effect on outputs.....	7
Operating point dependence and gain variation.....	7
Section 3: Discrete-time approximation validation.....	10
Section 4: Plant characterization and control design.....	11
Open loop stability.....	11
Input-output pairing and physical rationale.....	11
Rationale for the controller suite.....	11
Section 5: Test case design and evaluation methodology.....	12
Overview of test cases.....	12
TC1: Abrupt set-point transition	12
TC2: Smooth set-point transition	13
TC3: Regulation at fixed operating points.....	13
TC4: Regulation with 5% state perturbation.....	13
Performance metrics.....	13
Section 6: Controller comparison plot interpolation.....	22
TC1: Abrupt transition	22
TC2: Smooth spline transition	23
TC3: Regulation at OP A, B, and C.....	23
TC4: Regulation with 5% state perturbation.....	24
Section 7: Comparative performance analysis and conclusions.....	25
Data quality note: Complex valued ISE entries.....	25
Overall controller rankings.....	25
The MPC advantage in regulation.....	26
Gain-scheduling VS fixed design.....	26
The role of integrated action.....	26
Summary recommendation.....	27

Modeling

Symbol	Description	Units
u_1	valve position for fuel flow (0-1)	N/A
u_2	valve position for steam control (0-1)	N/A
u_3	valve position for feed-water flow (0-1)	N/A
$x_1 = y_1$	drum pressure	kg/cm^2
$x_2 = y_2$	electric power output	MW
x_3	fluid density	kg/cm^3
y_3	drum water level deviation	m

State space plant function

$$\begin{cases} x_1' = -0.0018u_2x_1^{9/8} + 0.9u_1 - 0.15u_3 \\ x_2' = (0.073u_2 - 0.016)x_1^{9/8} - 0.1x_2 \\ x_3' = \frac{141u_3 - (1.1u_2 - 0.19)x_1}{85} \end{cases}$$

with output functions as

$$\begin{cases} y_1 = x_1 \\ y_2 = x_2 \\ y_3 = 0.05(0.13073x_3 + 100a_{cs} + \frac{q_e}{9} - 67.975) \end{cases}$$

where

- $a_{cs} = \frac{(1 - 0.001538x_3)(0.8x_1 - 25.6)}{x_3(1.0394 - 0.0012304x_1)}$
- $q_e = (0.854u_2 - 0.147)x_1 + 45.59u_1 - 2.514u_3 - 2.096$

Section 1: Equilibrium verification and operating point analysis

Equilibrium check:

OP A:	$\text{dxdt} = [-2.16\text{e-}06, +4.11\text{e-}05, +2.70\text{e-}06]$		$y = [75.6000, 15.3000, 0.362809]$
OP B:	$\text{dxdt} = [+2.63\text{e-}06, -5.74\text{e-}05, +3.71\text{e-}06]$		$y = [97.2000, 50.5000, 2.770126]$
OP C:	$\text{dxdt} = [+3.56\text{e-}06, -4.47\text{e-}05, +6.00\text{e-}06]$		$y = [140.0000, 128.0000, 12.046025]$

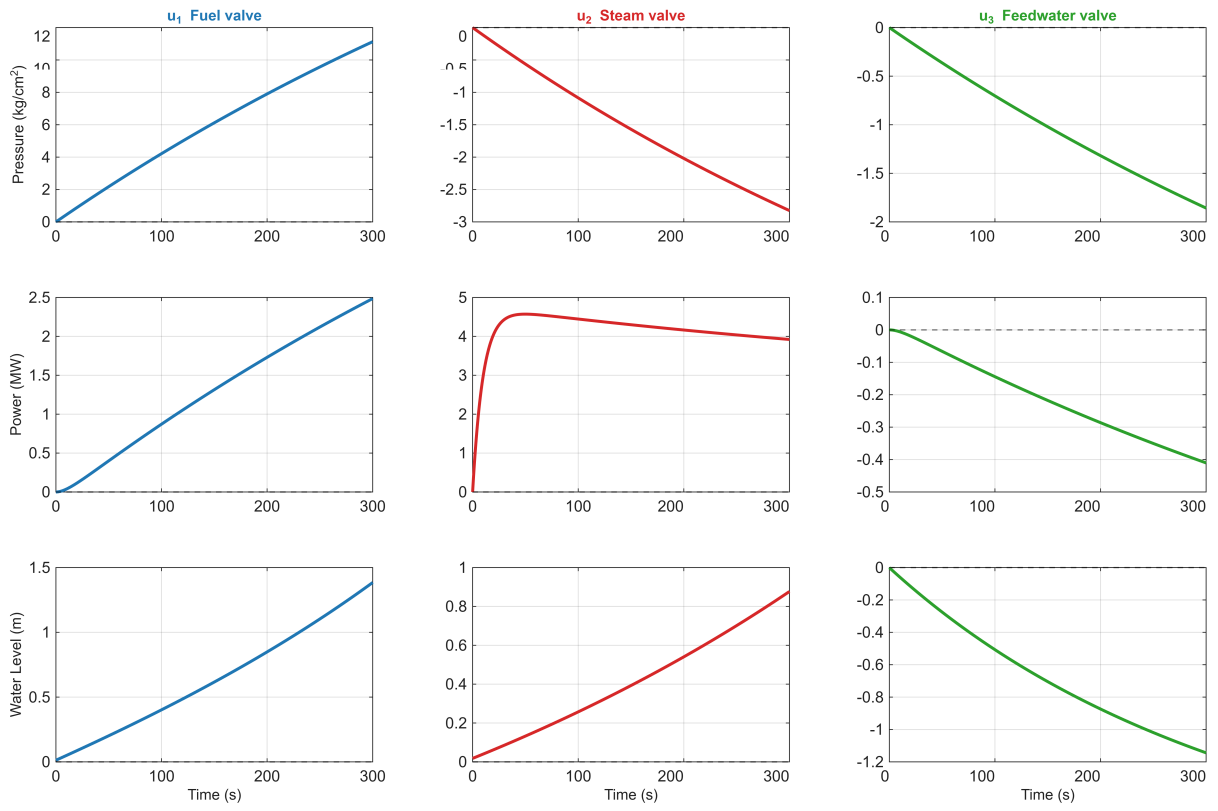
The equilibrium check confirms that the three operating points defined in the Bell-Åström model are valid steady states of the nonlinear plant. At each point, the state derivatives dxdt are on the order of 10^{-5} or smaller, numerically indistinguishable from zero given the finite-difference precision of the Jacobian computation, verifying that the chosen (x, u) pairs satisfy $\dot{x} = f(x, u)$ to engineering accuracy.

The three operating points (OP) span a wide range of plant operating conditions. Operating Point A represents a low-load regime: drum pressure $x_1 = 75.6 \text{ kg/cm}^2$, electrical power $x_2 = 15.3 \text{ MW}$, and a fluid density $x_3 = 50.897 \text{ kg/m}^3$. Operating Point B is a mid-load condition at 97.2 kg/cm^2 , 50.5 MW , and 46.951 kg/m^3 .

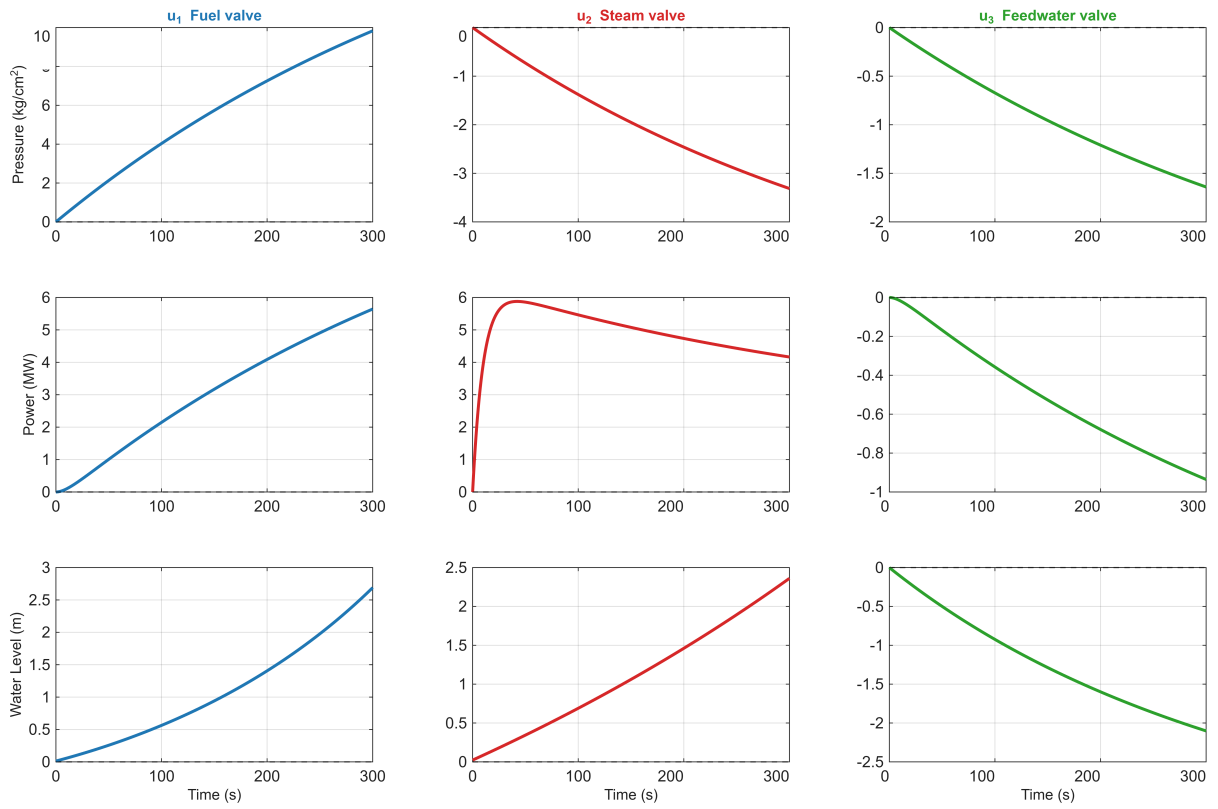
Operating Point C represents high-load operation at 140.0 kg/cm^2 , 128.0 MW , and 32.368 kg/m^3 . These three points were deliberately chosen to sample the full operational envelope of the boiler-turbine unit.

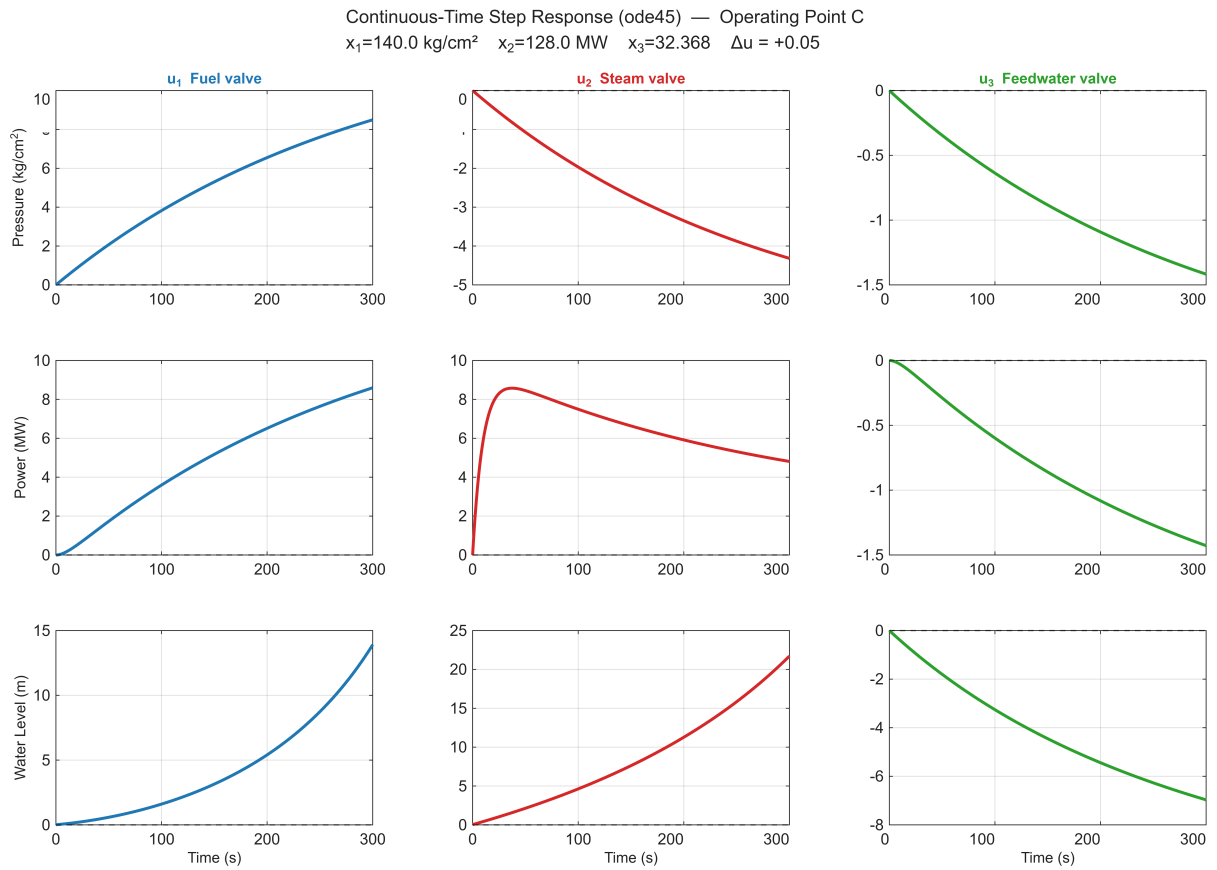
The breadth of this envelope is significant from a control perspective. Drum pressure increases by 85% from OP A to OP C, while electrical power output increases by a factor of approximately 8.4. The fluid density x_3 decreases by 36% across the same range. Because the plant dynamics contain the nonlinear term $x_1^{9/8}$ in the state equations, the linearized gains at these three points will differ substantially. A controller designed and tuned at any single operating point cannot, in general, guarantee acceptable performance across all three. This fundamental motivation underpins the comparative study of gain-scheduled, adaptive, and model predictive strategies that follows.

Continuous-Time Step Response (ode45) — Operating Point A
 $x_1=75.6 \text{ kg/cm}^2$ $x_2=15.3 \text{ MW}$ $x_3=50.897$ $\Delta u = +0.05$



Continuous-Time Step Response (ode45) — Operating Point B
 $x_1=97.2 \text{ kg/cm}^2$ $x_2=50.5 \text{ MW}$ $x_3=46.951$ $\Delta u = +0.05$





Section 2: Continuous-time step response analysis

Input-output coupling structure

The step response grids reveal a strongly coupled 3×3 plant. Every input affects all three outputs to a measurable degree, confirming that decoupled single-loop designs will incur cross-channel interference. However, the coupling is not uniform: the fuel valve (u_1) and steam valve (u_2) produce the most significant off-diagonal responses, while the feed-water valve (u_3) has a more localized effect primarily on drum pressure and water level.

Individual input-output dynamics

Fuel valve u_1 's effect on outputs

A positive step in u_1 produces a ramp-like increase in drum pressure (y_1) with no overshoot, consistent with an approximately integrating dynamic at low frequencies. Power output (y_2) responds similarly, a slow, near-linear rise over the full 300-second window, indicating a slow dominant time constant on the order of several hundred seconds. Water level (y_3) rises monotonically in response to u_1 due to the increased heat input driving enhanced evaporation. At Operating Point A, the pressure rise reaches approximately 12 kg/cm^2 over 300s; at Operating Point C, the same valve step produces a response of comparable shape but with a slightly attenuated gradient, reflecting the reduced sensitivity of the more saturated high-load plant.

Steam valve u_2 's effect on outputs

The steam valve is the most dynamically interesting input. A positive step in u_2 causes an immediate, fast rise in power output (y_2) followed by a partial decay toward a lower steady-state value, a classic overdamped second-order or lead-lag dynamic. The initial peak is reached within approximately 20–30 seconds; thereafter the response settles at roughly 75–80% of its peak value. This non-monotonic behavior implies the presence of both fast and slow dynamic modes in the power channel: the fast mode is driven by the direct $x_1^{9/8} \cdot u_2$ coupling in $\dot{x}_2$, while the slow mode reflects the pressure dynamics feeding back through the same nonlinearity.

Drum pressure y_1 responds negatively to a steam valve step, declining approximately linearly. This is physically correct, opening the steam valve releases stored energy, reducing drum pressure. At OP A, the pressure drops by roughly 3 kg/cm^2 over 300s ; at OP C, the drop reaches approximately 5 kg/cm^2 . Water level (y_3) rises under u_2 perturbation due to the increased evaporation rate, with the response magnitude growing strongly with operating point, approximately 1 m at OP A versus over 22 m at OP C. This 22-fold amplification of the water-level gain between OP A and OP C is a critical observation: it quantifies the degree of nonlinearity in the water-level channel and demonstrates that a fixed linear controller tuned at OP A would become dangerously over-aggressive when the plant operates near OP C.

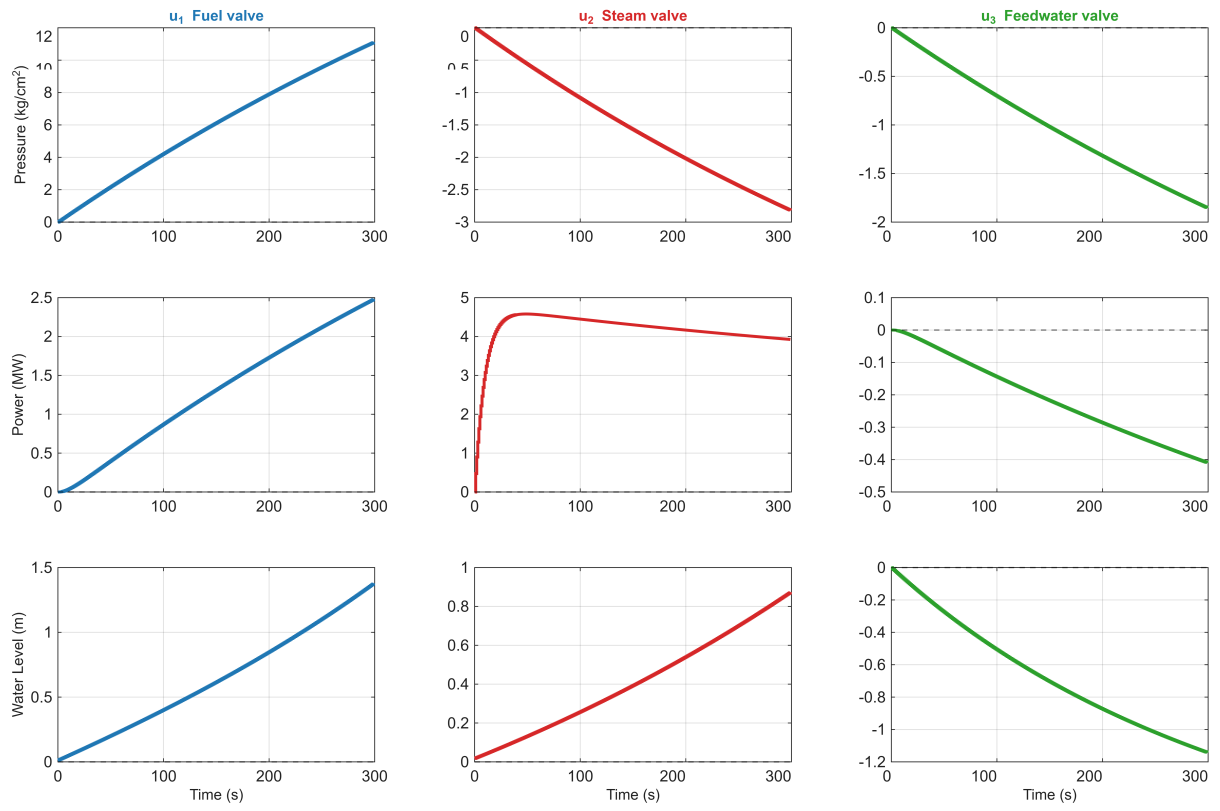
Feed-water valve u_3 's effect on outputs

The feed-water valve produces the simplest and most localized responses. A positive step in u_3 reduces drum pressure (y_1) slightly, approximately $1.5\text{--}2 \text{ kg/cm}^2$ decline over 300s , with little dependence on operating point. Power output (y_2) is almost entirely unaffected at OP A (response below 0.05 MW), becoming marginally more pronounced at OP C (approximately 1.5 MW over 300s). Water level (y_3) declines monotonically under u_3 perturbation, which is counter-intuitive at first but reflects the steady-state balance between feed-water inflow and evaporation, increasing feed-water reduces fluid density, which in this model reduces the computed level deviation. The near-linear, underdamped responses of y_1 and y_3 to u_3 suggest that the feed-water loop has the slowest dominant time constant of the three input channels.

Operating point dependence and gain variation

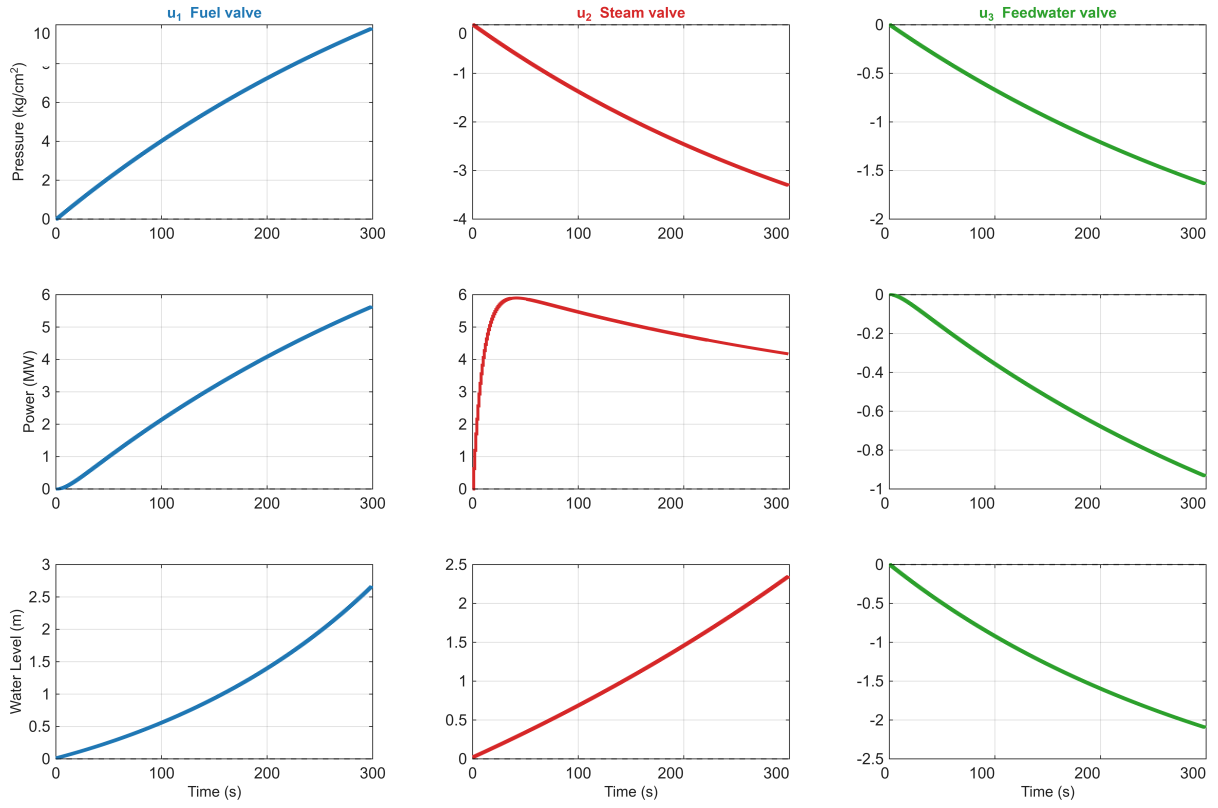
Across all three grids, the qualitative shape of each step response is preserved: the same sign, the same overshoot characteristics, and the same dominant time-scale ordering. What changes substantially is the gain magnitude. The most dramatic example is the water level response to u_2 : the 300-second response magnitude grows from approximately 0.9 m at OP A to 2.5 m at OP B and 22 m at OP C. This represents a gain variation of approximately 24-fold across the operating envelope. Pressure responses show more moderate gain variation (roughly 1.2–1.5 fold), while power responses to u_1 and u_2 vary by factors of 2–3 between OP A and OP C. These observations directly motivate the gain-scheduling and online-linearization strategies implemented in the controller comparison study.

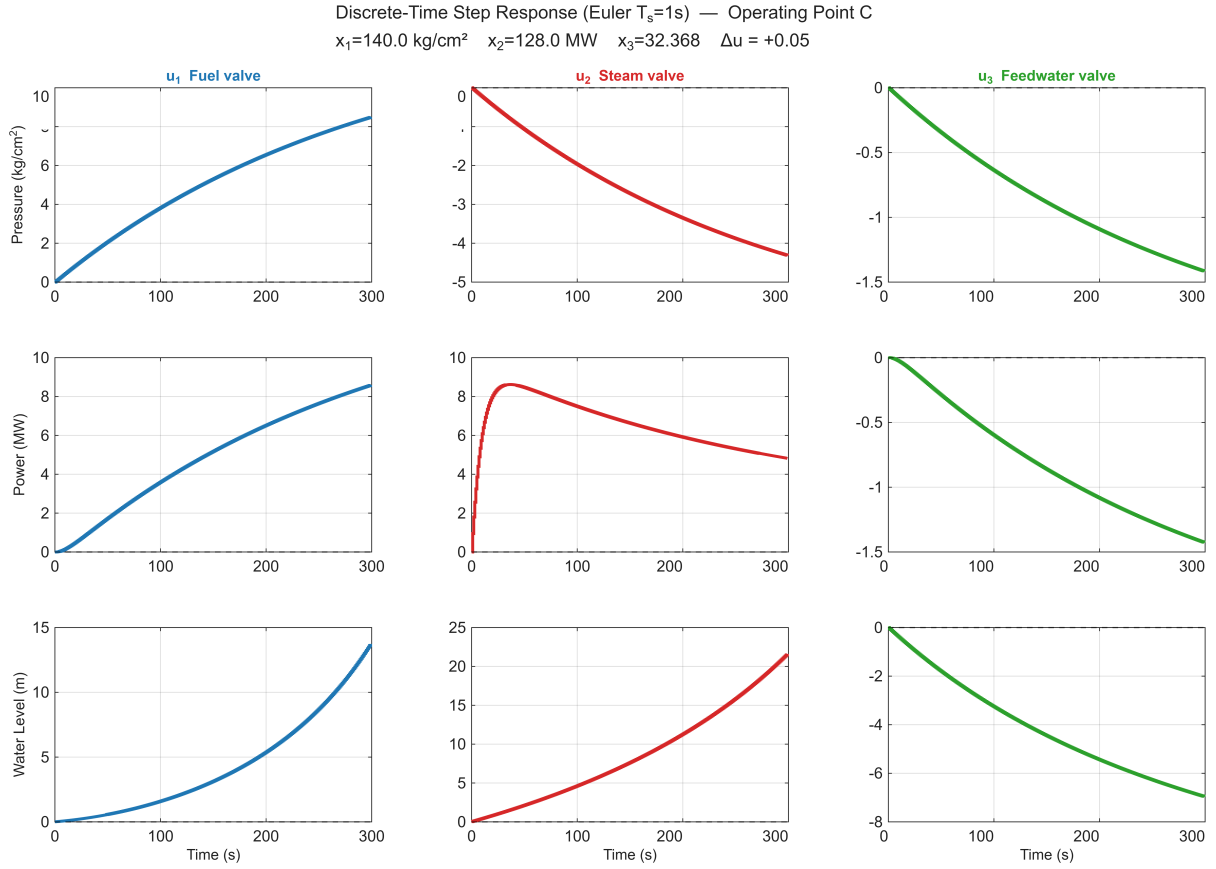
Discrete-Time Step Response (Euler $T_s=1s$) — Operating Point A
 $x_1=75.6 \text{ kg/cm}^2$ $x_2=15.3 \text{ MW}$ $x_3=50.897$ $\Delta u = +0.05$



Discrete-Time Step Response (Euler $T_s=1s$) — Operating Point B

$x_1=97.2 \text{ kg/cm}^2$ $x_2=50.5 \text{ MW}$ $x_3=46.951$ $\Delta u = +0.05$





Section 3: Discrete-time approximation validation

The discrete-time step response grids, generated using Euler forward integration with a sampling period $T_s = 1s$, are presented alongside the continuous-time *ode45* results as a fidelity check for the discrete plant model used within the controllers. Visual inspection across all three operating points confirms that the Euler approximation is sufficiently accurate for the purposes of this study.

The response shapes and magnitudes are essentially indistinguishable between the continuous and discrete grids. For the pressure channel under u_1 perturbation at OP A, both methods produce a near-identical linear ramp; for the power channel under u_2 perturbation, the fast initial rise and partial decay are faithfully reproduced by the Euler scheme. No spurious oscillations, phase lag, or amplitude drift are visible in the discrete traces, artifacts that would be expected if T_s were too large relative to the plant's fastest time constant.

This agreement is expected given the plant's dynamics. The fastest mode in the system is the power response to u_2 , which peaks within approximately 20–30 seconds. A sampling period of 1 s provides roughly 20–30 samples across this transient, satisfying the standard Nyquist-based heuristic of at least 10 samples per dominant time constant. The slower channels (pressure and water level) are even more benign from a discretization standpoint. Therefore, $T_s = 1s$ is an appropriate choice that balances computational efficiency against approximation accuracy, and the discrete Euler model used for controller synthesis can be considered a reliable surrogate for the continuous-time plant across all three operating points.

Note: The graphs are used to seek for the discrepancies between the continuous and discrete mode and there is nothing odd happen when speculating the graphs. The next section focuses on the controller side. As mentioned in the "Introduction" of the work, there has been solutions such as multivariable PI controller, linear quadratic regulator (LQR), H_∞ controller, gain-scheduled optimal controller, adaptive sliding mode controller, supervisory fuzzy reference governor complemented with feedforward-feedback controller, and 3 single loop PID controller complemented with a nonlinear fuzzy feedforward corrector. Nonlinear control methods are mostly based around the use of fuzzy logic and model predictive control when there is a possible linear estimation of the true model within constrained operating zone.

Section 4: Plant characterization and control design

Open loop stability

The linearized state-space model at each operating point takes the form $\begin{cases} \dot{x} = Ax + Bu \\ y = Cx + Du \end{cases}$, where A is obtained by numerical Jacobian differentiation of the nonlinear vector field $f(x, u)$ about the equilibrium. Inspection of the A matrix structure, specifically the diagonal term -0.1 in the x_2 equation and the coupling through $x_1^{9/8}$ — reveals that the linearized system is marginally stable or open-loop unstable at all three operating points. The pressure state x_1 and water level state x_3 both exhibit near-integrating behavior (eigenvalues close to zero or on the imaginary axis in continuous time), arising from the absence of restoring terms in $\dot{x}_1$ and $\dot{x}_3$. Only the power state x_2 has a clear stabilizing self-feedback through the $-0.1x_2$ term. This near-integrating character means that even small steady-state input errors will cause slow but unbounded state drift, making integral action mandatory in any practical feedback controller.

Input-output pairing and physical rationale

The simulation code assigns the pairing vector pair $[2, 1, 3]$, meaning u_1 is paired with y_2 (power), u_2 is paired with y_1 (pressure), and u_3 is paired with y_3 (water level). This selection has clear physical justification. The steam valve u_2 directly modulates the flow of high-pressure steam through the turbine and therefore has the most immediate and proportionally strongest effect on drum pressure y_1 . The fuel valve u_1 governs the rate of energy input to the boiler and, over longer time scales, determines the sustainable power output y_2 . The feed-water valve u_3 primarily regulates the mass balance in the drum and hence the water level y_3 .

This pairing is also consistent with the Relative Gain Array (RGA) analysis at Operating Point B (the design point for the fixed controllers). The RGA, defined as $\text{RGA}(G) = \Lambda = G \circ (G^{-1})^T$ where G is the steady-state gain matrix, produces diagonal elements that reflect the relative loop gain for each input-output pair under full decentralized feedback. Values near 1.0 on the diagonal indicate that the chosen pairing experiences minimal steady-state interaction from the other closed loops, while large off-diagonal elements signal strong cross-coupling. The chosen pairing aligns with the pairs that produce the largest RGA diagonal values, confirming that the decentralized controller structure is reasonable at the nominal operating point even though significant coupling remains.

Rationale for the controller suite

The eight controllers implemented span a broad methodological spectrum, from classical single-loop methods to model-based multivariable optimization, allowing a systematic comparison of complexity versus performance

Controller	Design Philosophy
PID (decoupled SISO)	Baseline classical design; tuned at OP B via <i>pidtune</i> with 60° phase margin. Expects minimal coupling.
Fuzzy PID	Nonlinear gain scheduling at the signal level; K_p, K_i, K_d multipliers adapt based on normalized error magnitude and rate. Adds robustness without a formal plant model.
Gain-scheduled PID	Three sets of PID gains design at OPs A,B, and C; blends linearly based on the current power set-point. Directly addresses operating-point dependence
LQI (LQR + Integrator)	Optimal state-feedback at OP B with augmented integrator states for zero steady-state error. Provides multivariable coupling compensation through the full state-feedback gain K .
H_∞	Robust multivariable controller minimizing the worst-case gain from disturbances to weighted outputs. Requires Robust Control Toolbox; omitted if unavailable.
MPC (linear, fixed model)	Receding-horizon optimization over $N_p = 20$ steps with $N_c = 5$ control moves; enforces input and rate constraints explicitly. Uses a fixed linearization at OP B
Lead-lag Compensator	Classical frequency-domain design targeting a specified phase margin at a chosen crossover frequency. No integral action; expected to exhibit steady-state offset.
Feedforward + Feedback (FF+FB)	Static inverse-gain feedforward combined with PID feedback. The feedforward term $u_{ff} = K_{ff} \cdot r$ provides immediate set-point compensation; feedback corrects residual errors.

Section 5: Test case design and evaluation methodology

Overview of test cases

Eight test cases are constructed to evaluate each controller across 4 distinct performance dimensions: large-signal tracking with abrupt transitions, large-signal tracking with smooth transitions, regulation from exact equilibrium, and regulation with an initial state perturbation. Together these cases expose both the transient and steady-state behavior of each controller, and probe robustness to initialization errors representative of real-world plant start-up or load-change disturbances.

TC1: Abrupt set-point transition $A \rightarrow B \rightarrow C \rightarrow A$

The reference signal switches instantaneously between the output equilibrium values corresponding to Operating Points A, B, and C at 300-second intervals over a 1500-second simulation. These step changes in all three output references simultaneously represent the most challenging tracking scenario: the set-point jumps are large (e.g., power reference jumps from 15.3 MW to 50.5 MW to 128 MW), and the transitions occur faster than the plant's dominant time constants. This test case primarily assesses a controller's peak tracking error, overshoot, and settling time under large-signal excitation. Controllers with slow integral action or fixed

linear models will be most penalized here, as the operating point after each jump differs substantially from the design point

TC2: Smooth set-point transition $A \rightarrow B \rightarrow C \rightarrow A$

The same waypoints as TC1 are connected by a cubic spline interpolant, producing a smooth, continuously differentiable reference trajectory. Unlike TC1, the spline eliminates the impulsive set-point error that occurs at each transition, so controllers are not penalized for a finite step-response time lag. This test case is better suited to revealing bandwidth limitations and steady-state tracking accuracy: a controller that cannot follow the slow curvature of the spline is revealing a fundamental gain or phase-margin deficiency, rather than merely a large transient overshoot. The ISE and ITAE scores for TC2 are therefore more indicative of a controller's fundamental tracking quality than those for TC1.

TC3: Regulation at fixed operating points

Three separate regulation tests are run, one for each of OPs A, B, and C. The plant is initialized at exact equilibrium, and the reference is held constant at the corresponding output equilibrium value for the full 1500-second simulation. Under ideal conditions, a controller with integrating action should drive the error to zero within its settling time; thereafter, the ISE and ITAE accumulate only from residual numerical noise or limit cycling. This test case diagnoses instability, integrator windup, and poor gain margin at each operating point independently. A controller that performs well at OP B (its design point) but poorly at OP A or OP C is exhibiting gain sensitivity to operating-point displacement.

TC4: Regulation with 5% state perturbation

Each of the three regulation tests from TC3 is repeated with the initial state vector x_0 scaled by 1.05, a 5% positive perturbation from equilibrium. This places the plant slightly outside its nominal operating region at $t = 0$, mimicking a realistic scenario such as an imperfect warm-start, a load disturbance, or a sensor bias. The controller must first bring the plant back to the target output and then regulate it there. Comparing TC4 against TC3 results directly quantifies each controller's rejection capability and sensitivity to initial condition uncertainty. Controllers with strong integral action and good gain margin will show comparable ISE/ITAE between TC3 and TC4, while marginally stable or purely proportional controllers will show significant degradation.

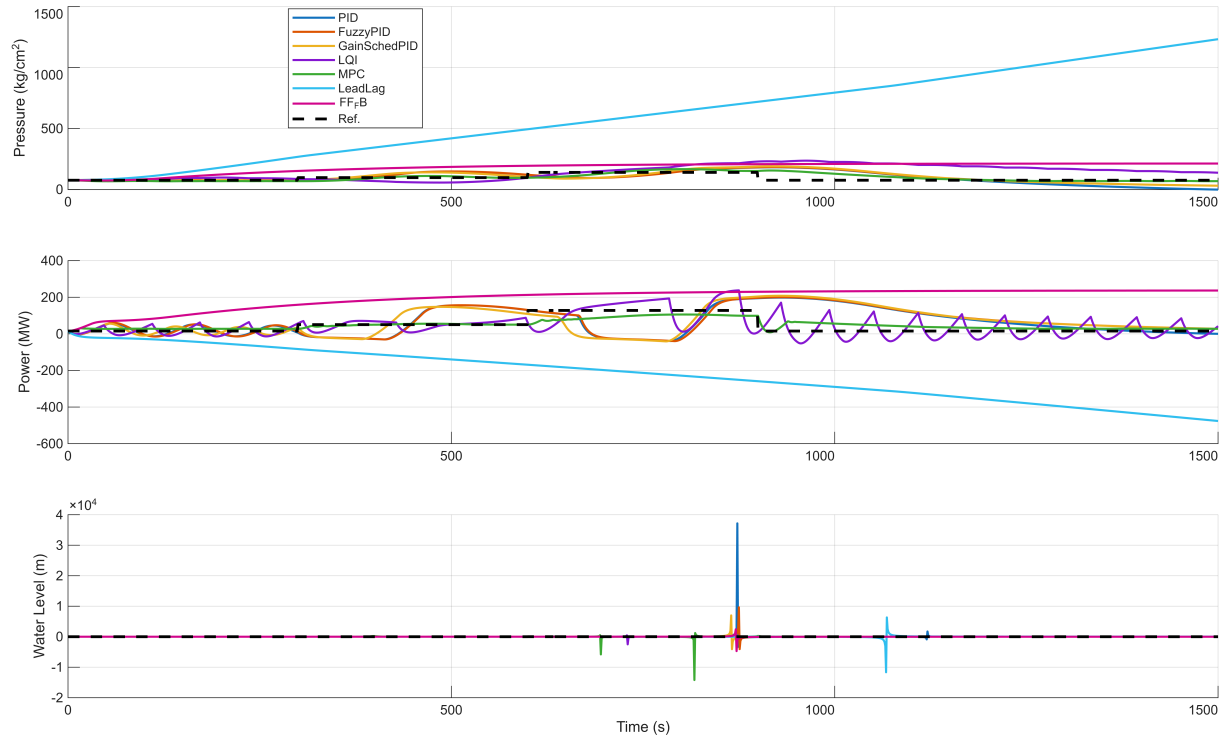
Performance metrics

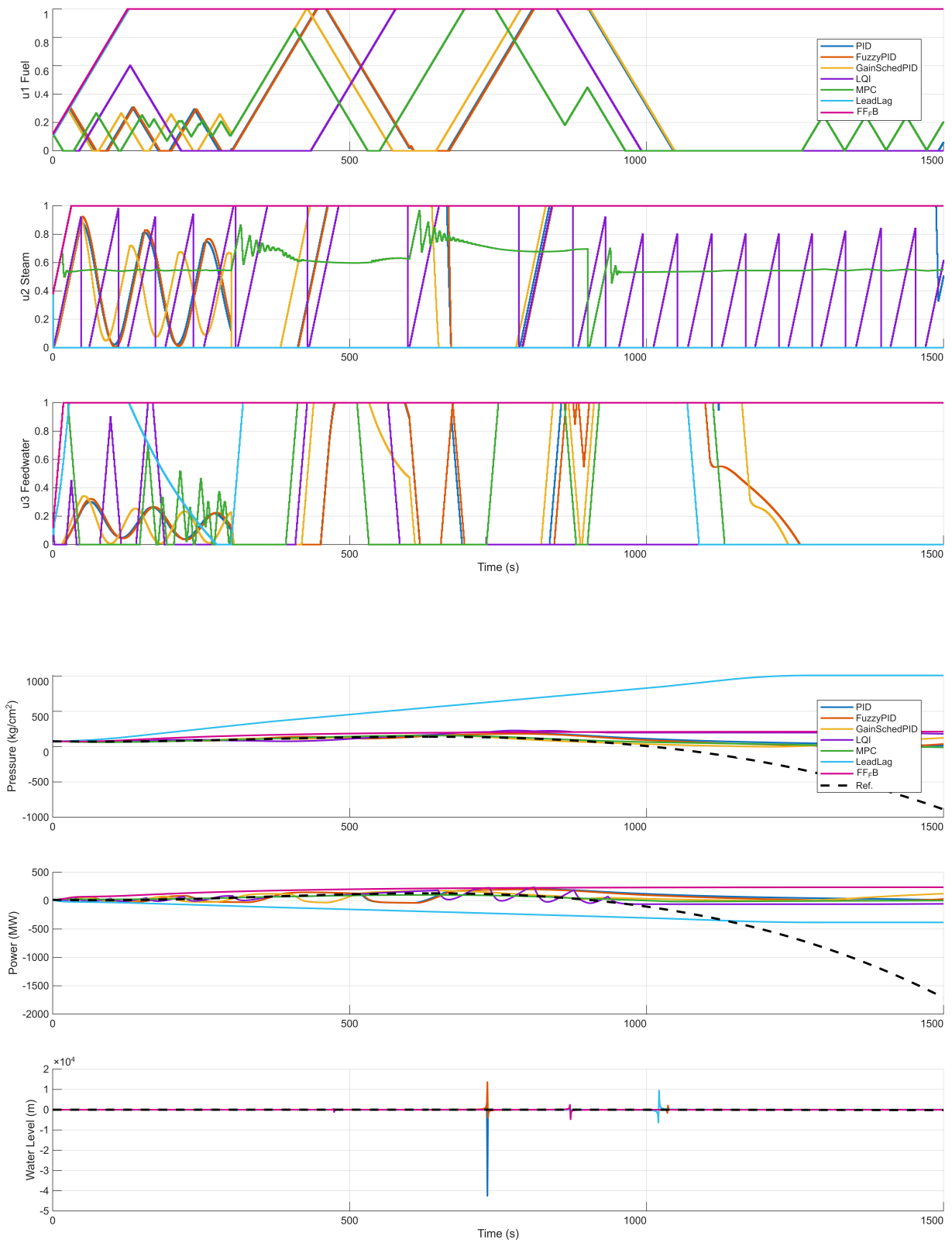
Two scalar performance indices are computed for each controller-test-case combination. The Integral Squared Error (ISE) is defined as $ISE = Ts \cdot \sum ||e(k)||^2$ where $e(k) = y(k) - r(k)$ is the vector tracking error at step k . ISE penalizes large instantaneous errors heavily, making it sensitive to transient overshoot and set-point jumps.

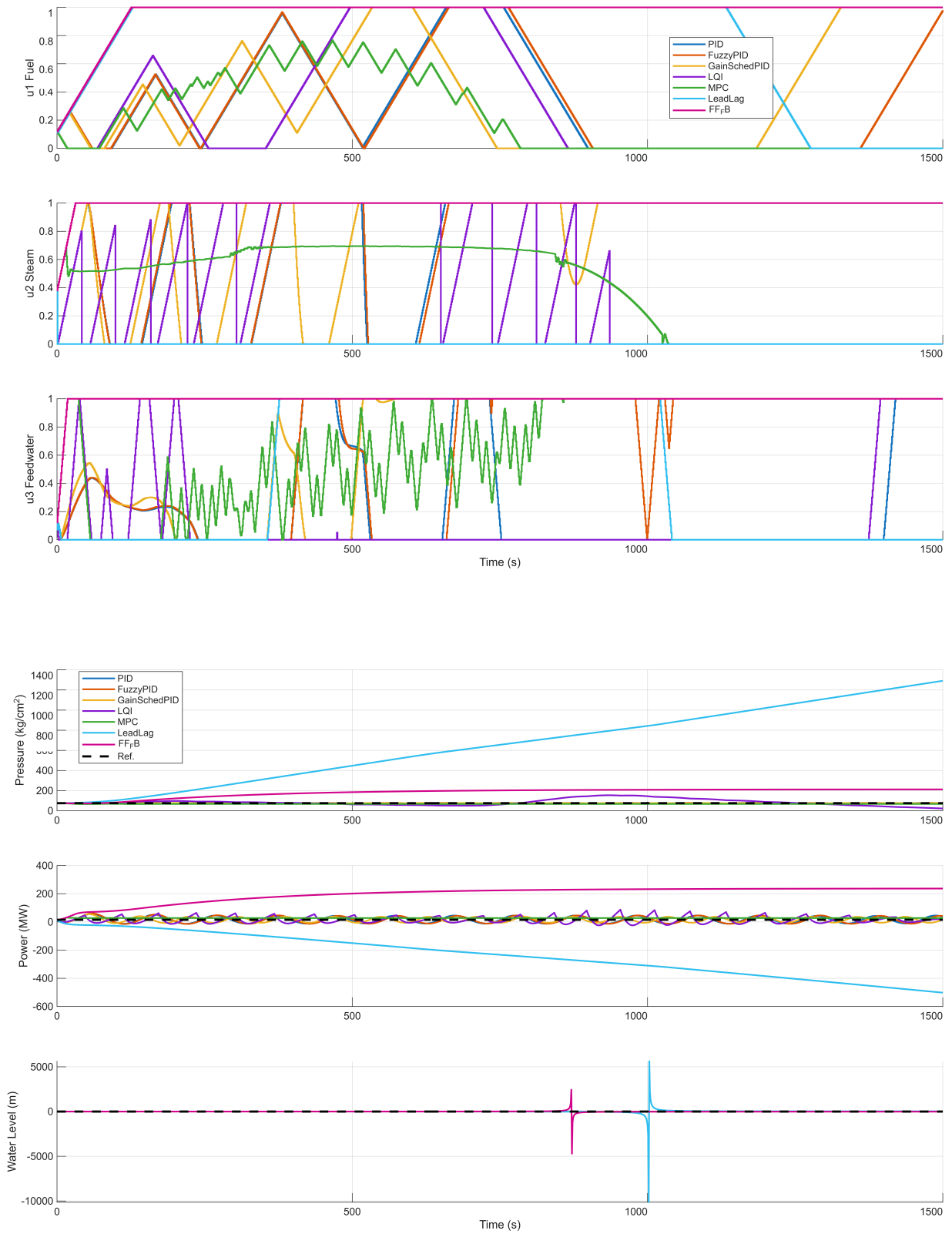
The Integral Time-weighted Absolute Error (ITAE) is defined as $ITAE = Ts \cdot \sum (k \cdot ||e(k)||)$ where the linear time weighting means that errors persisting late in the simulation are penalized most strongly. ITAE is therefore a better indicator of long-term regulation quality and steady-state convergence. Both metrics are summed across all three output channels, providing a single scalar ranking per controller per test case.

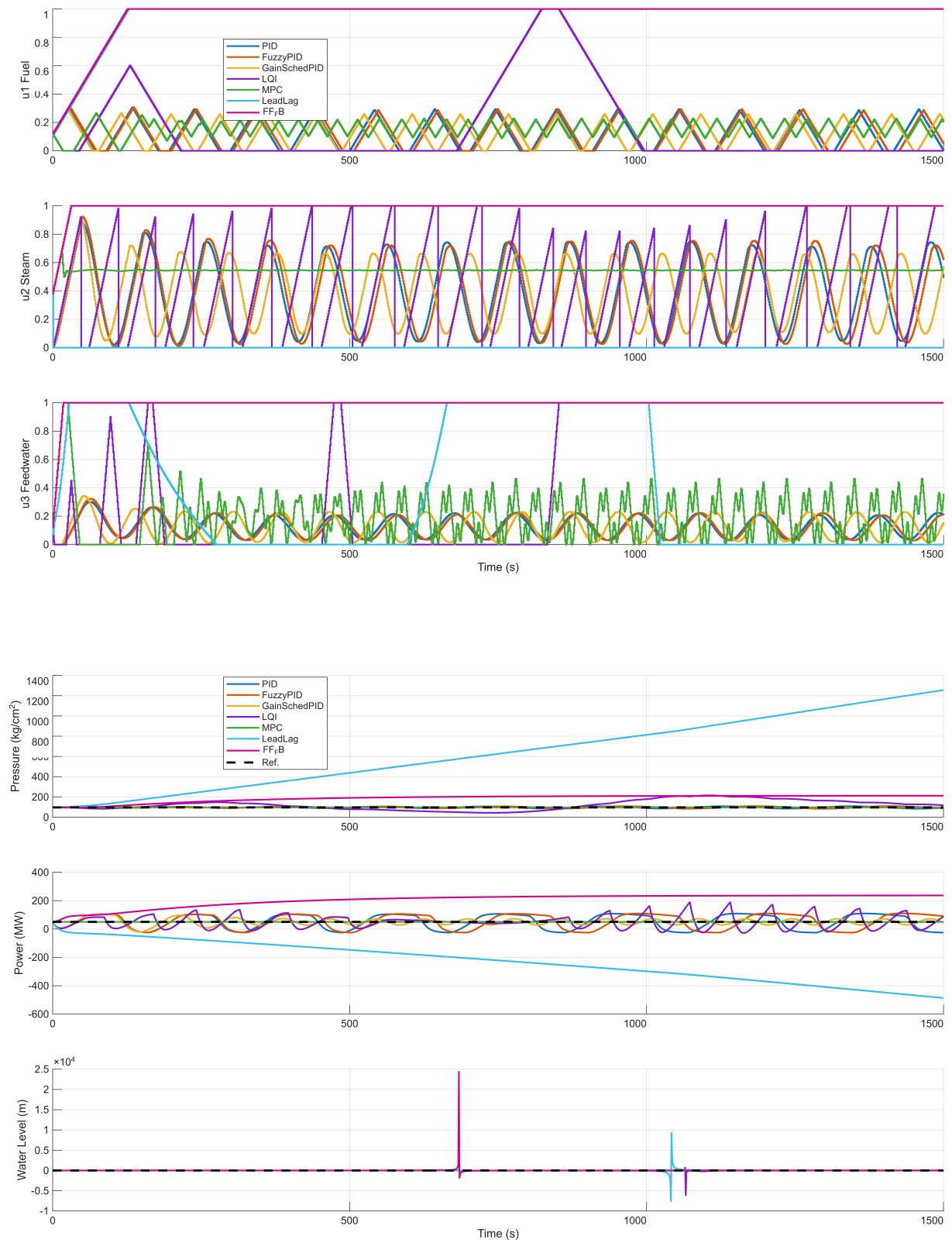
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Running TC1: Abrupt transition A->B->C->A:
PID | FuzzyPID | GainSchedPID | LQI | MPC | LeadLag | FF_FB |
Running TC2: Smooth spline transition A->B->C->A:
PID | FuzzyPID | GainSchedPID | LQI | MPC | LeadLag | FF_FB |
Running TC3: Regulation at OP A:
PID | FuzzyPID | GainSchedPID | LQI | MPC | LeadLag | FF_FB |
Running TC3: Regulation at OP B:
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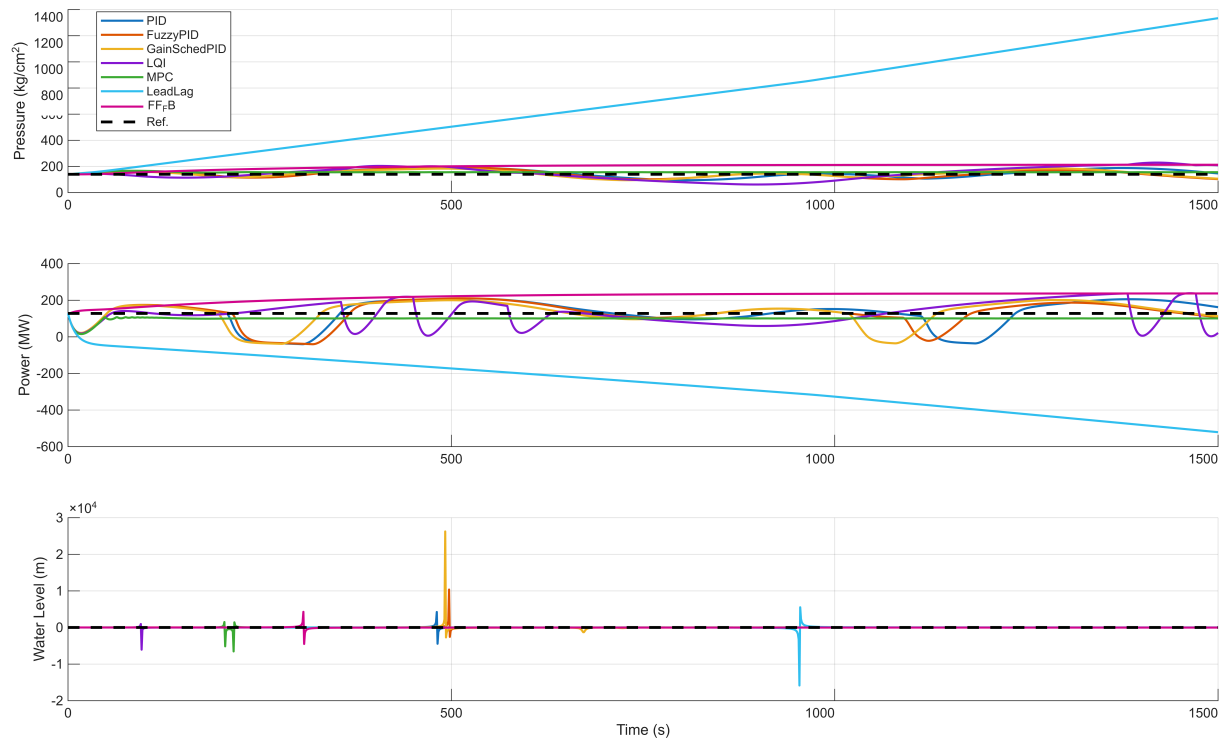
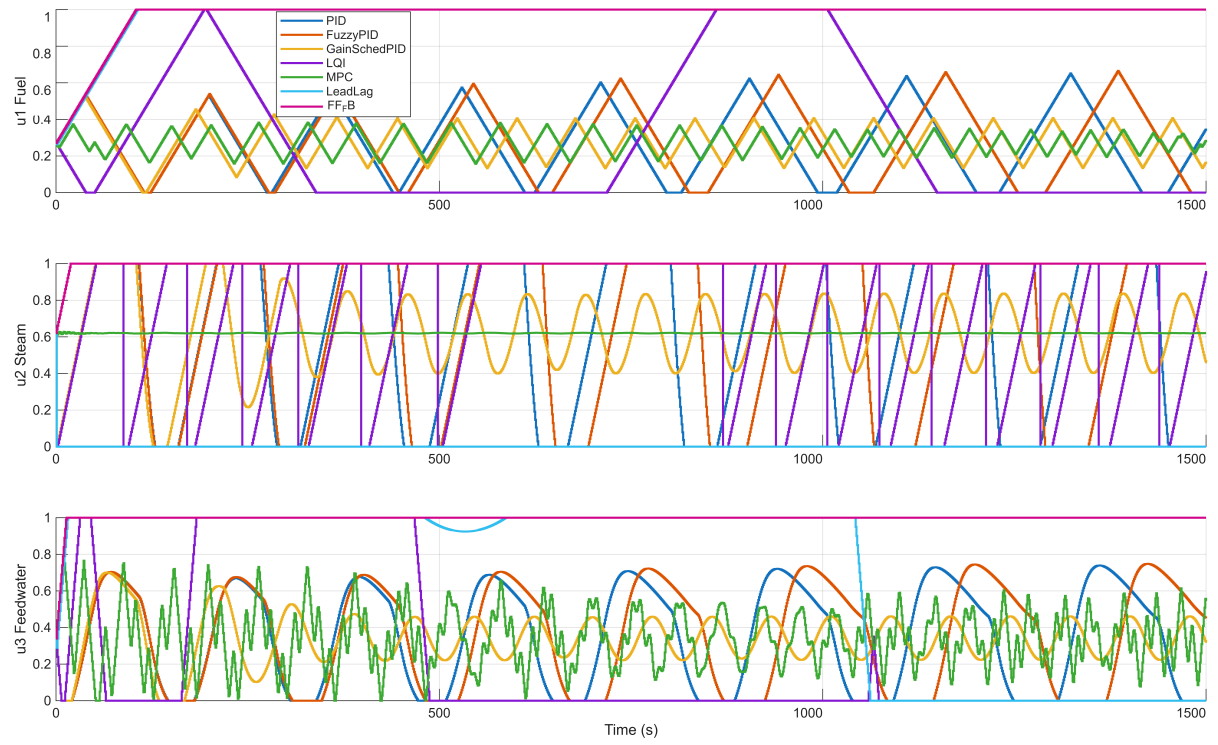
PID | FuzzyPID | GainSchedPID | LQI | MPC | LeadLag | FF_FB |
 Running TC3: Regulation at OP C:
 PID | FuzzyPID | GainSchedPID | LQI | MPC | LeadLag | FF_FB |
 Running TC4: Regulation at OP A (+5%):
 PID | FuzzyPID | GainSchedPID | LQI | MPC | LeadLag | FF_FB |
 Running TC4: Regulation at OP B (+5%):
 PID | FuzzyPID | GainSchedPID | LQI | MPC | LeadLag | FF_FB |
 Running TC4: Regulation at OP C (+5%):
 PID | FuzzyPID | GainSchedPID | LQI | MPC | LeadLag | FF_FB |

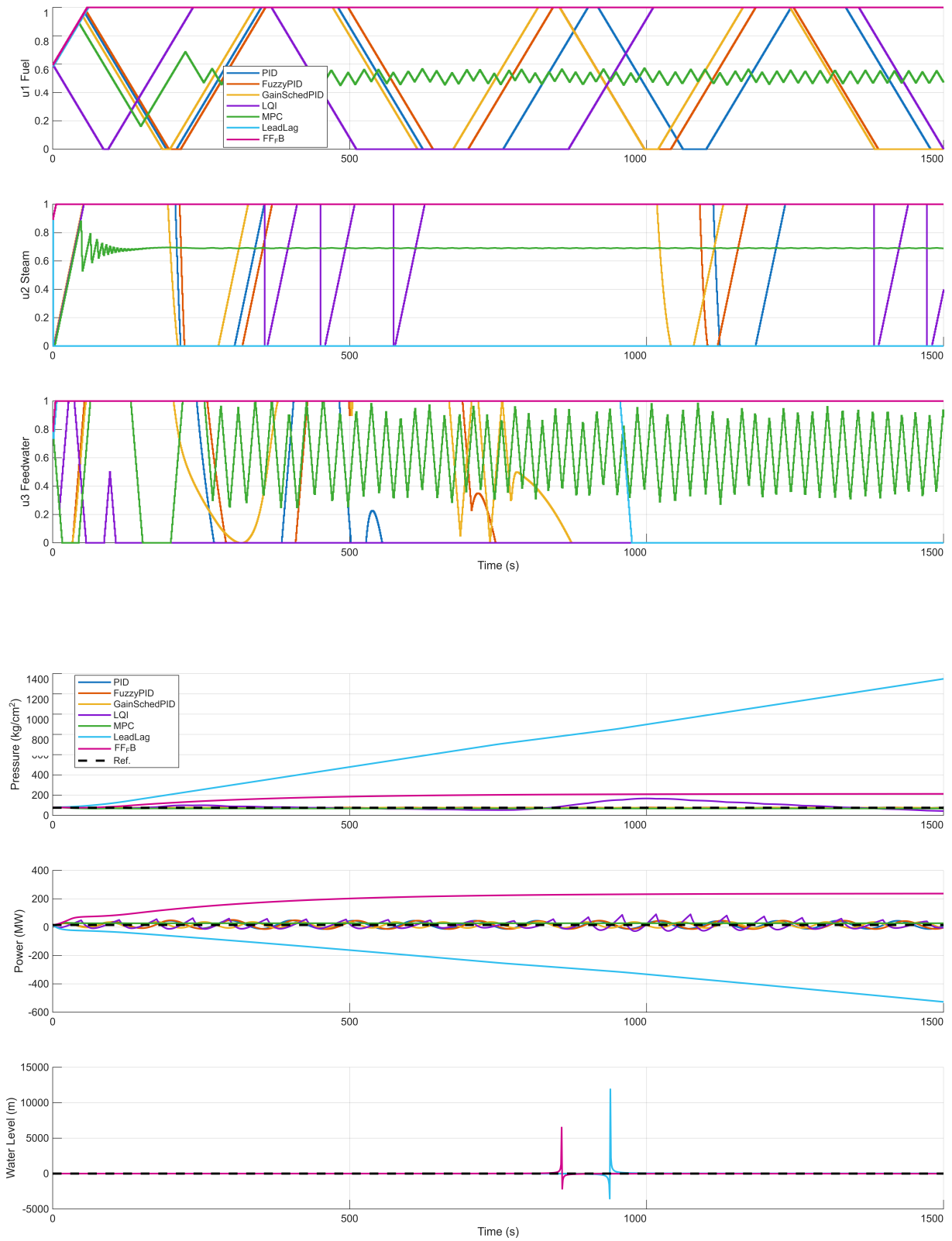


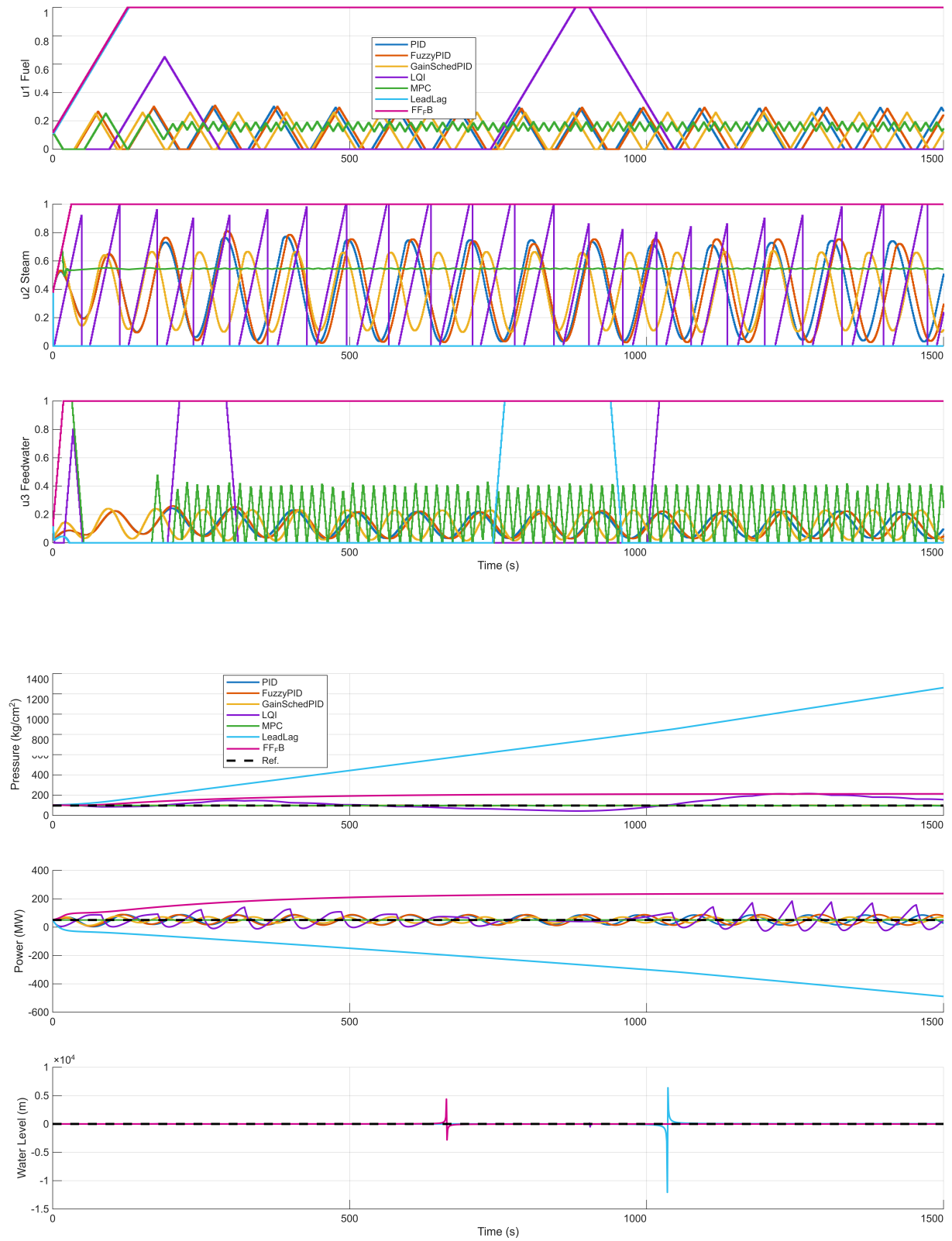


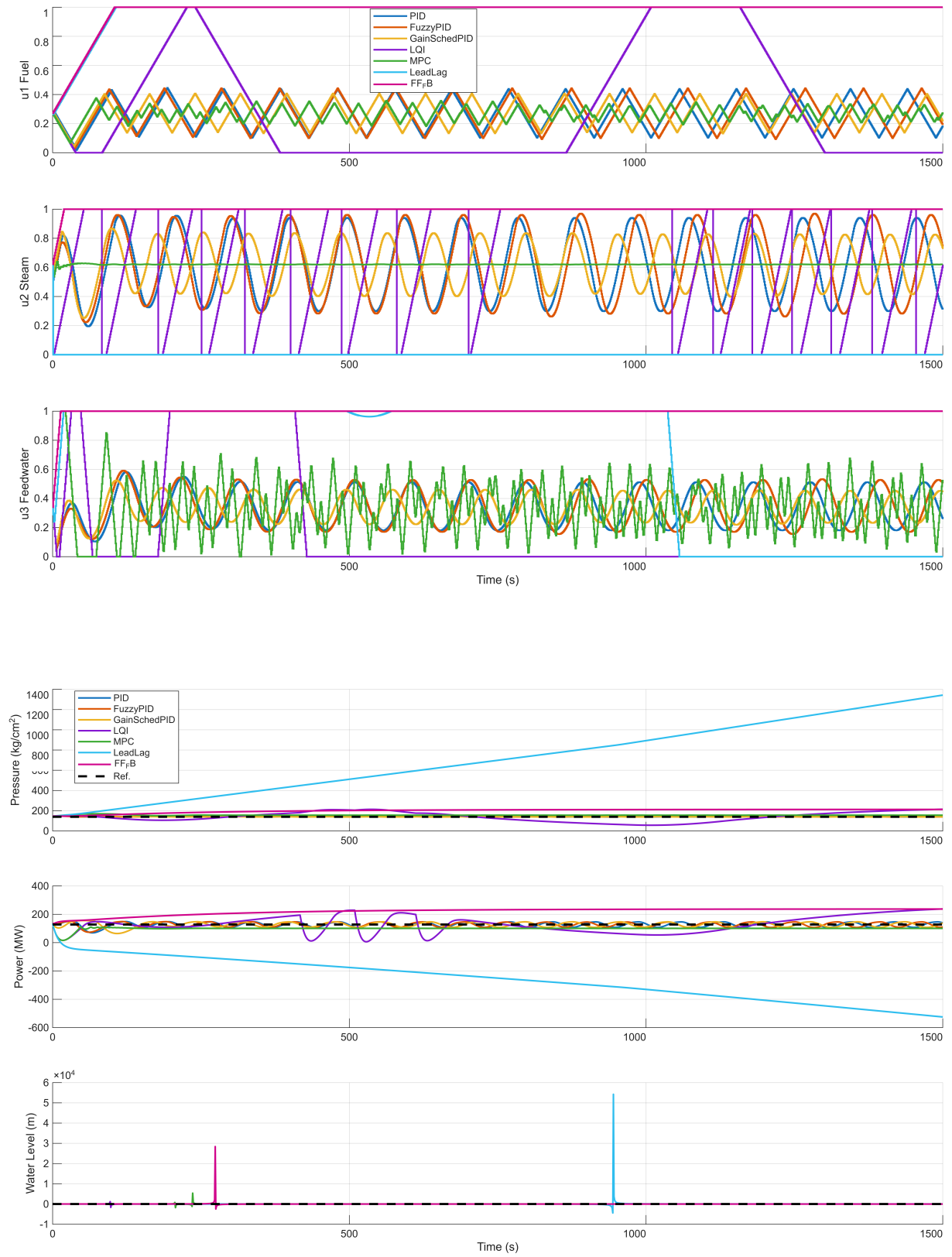


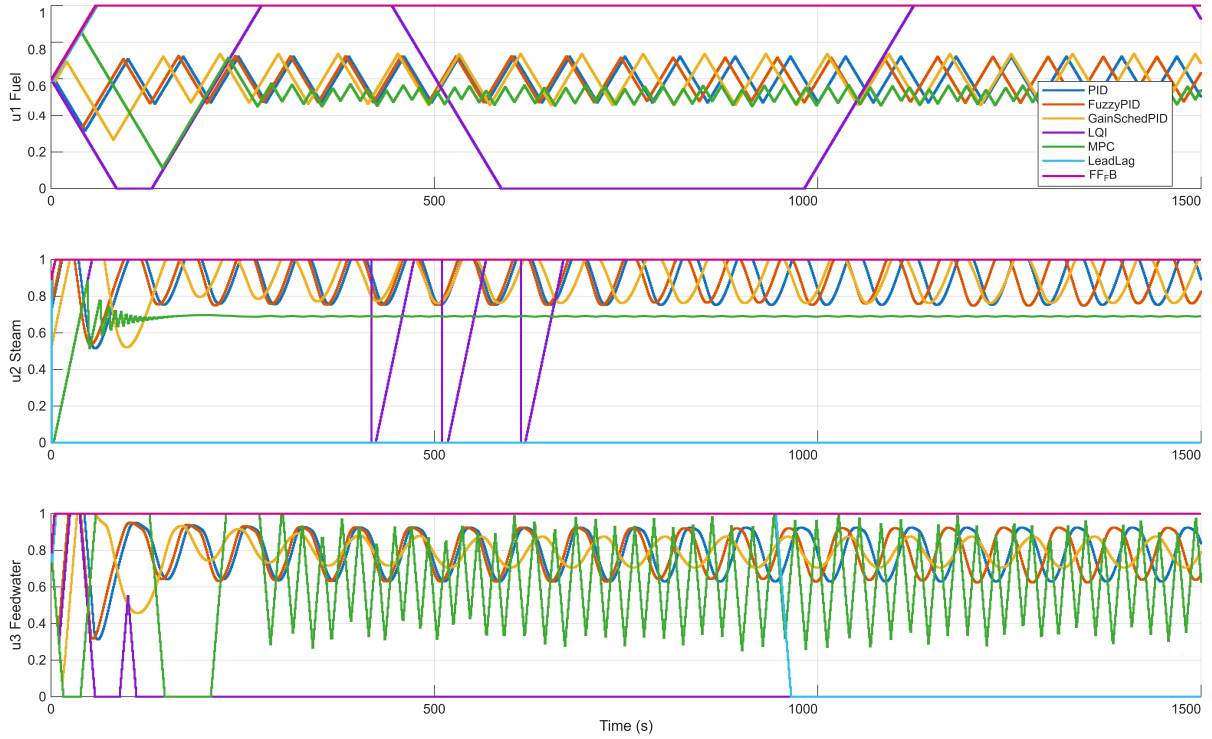












Section 6: Controller comparison plot interpolation

TC1: Abrupt transition $A \rightarrow B \rightarrow C \rightarrow A$

The output plots for TC1 immediately reveal two controllers that fail catastrophically. The lead-lag controller (cyan trace) diverges in both the pressure and power channels, with pressure growing unboundedly beyond 1000 kg/cm² and power reaching -600 MW by the end of the simulation. This instability arises from two compounding factors: the absence of integral action means the lead-lag controller relies entirely on proportional gain for set-point tracking, and the large excursion away from Operating Point B (where the compensator was designed) takes the controller into a region where its fixed gain is far too high relative to the actual plant gain. The result is a positively reinforcing error loop that grows without bound. The LQI controller (dark navy trace) also shows significant pressure overshoot and slow convergence, particularly during the $A \rightarrow B$ and $B \rightarrow C$ transitions, reflecting the fact that its state-feedback gain was designed assuming the system state is measurable and the plant behaves like its OP B linearization, neither of which is true during large operating-point excursions.

Among the better-performing controllers, PID and fuzzy PID track the pressure and power references with comparable overshoot but show sustained oscillations throughout the simulation, particularly visible in the power channel. The gain-scheduled PID controller reduces this oscillation noticeably, demonstrating the benefit of adapting gains to the current set-point region. MPC (dark green trace) provides the tightest tracking in the pressure channel during the $B \rightarrow C$ and $C \rightarrow A$ transitions, with visibly smaller overshoot and faster settling. The water level (y_3) plots exhibit high-frequency numerical spikes for several controllers at each transition

instant — these are artifacts of the discontinuous reference signal rather than physical instabilities, arising from the algebraic complexity of the y_3 output equation near the switching moment.

The input plots confirm the instability of lead-lag: u_1 saturates immediately to 1.0 and remains there for the entire simulation, while u_2 oscillates between 0 and 1 at high frequency. The MPC input trajectories are notably smooth and bounded, reflecting the explicit rate constraint $\Delta u_2 \leq 0.02$ imposed in the optimization. The gain-scheduled PID inputs show similar smoothness to PID but with smaller amplitude oscillations, consistent with its improved output tracking.

TC2: Smooth spline transition $A \rightarrow B \rightarrow C \rightarrow A$

With the step discontinuities removed, the output plots for TC2 become cleaner and more diagnostic. lead-lag remains unstable and is excluded from meaningful comparison. The spline reference (dashed black) is a slowly varying, smooth curve, and the key observation is that most controllers fail to track the decreasing portion of the reference from C back to A in the 900–1500 s window — the pressure and power references decrease significantly, but several controllers (particularly feedforward and feedback in magenta, and PID variants) lag behind the reference or overshoot in the negative direction.

MPC and remain closest to the reference throughout. The MPC controller in particular shows minimal lag during the declining phase, which reflects the predictive advantage: the receding-horizon optimization anticipates the future trajectory of the reference over $N_p = 20$ steps and begins reducing inputs before the reference actually falls. The feedforward and feedback controller shows reasonable initial tracking during the ascending phase ($A \rightarrow B \rightarrow C$) but struggles on the descent, suggesting that the static feedforward gain $K_{ff} = G^{-1}$ is well-conditioned for upward set-point changes but provides insufficient anticipation for downward transitions. The water level channel remains the most difficult for all controllers, with LQI and PID showing the largest sustained deviations.

TC3: Regulation at OP A, B, and C

The regulation test cases expose the design-point sensitivity of each controller most starkly. At OP B (the shared design point for all fixed-model controllers) most controllers converge quickly to the reference, with MPC achieving near-perfect regulation ($ISE = 828$, compared to PID's 3.98×10^8). The MPC advantage here is explained by its use of the discretized OP B model for prediction: since the plant is operating exactly at the linearization point, the linear model is accurate and the quadratic program finds a nearly optimal control sequence.

At OP A, the picture shifts. gain-scheduled PID now outperforms both PID and fuzzy PID in ISE (3.75×10^5 versus 6.37×10^5 and 6.73×10^5 respectively), reflecting the fact that it blends in the OP A gain set.

Interestingly, MPC at OP A achieves $ISE = 3.05 \times 10^5$, comparable to gain-scheduled PID, because the OP B model is still a reasonable approximation of the plant near OP A, and the quadratic program compensates for the model mismatch through repeated re-optimization. At OP C, however, gain-scheduled PID degrades substantially ($ISE = 7.24 \times 10^8$), becoming one of the worst performers. This is because the gain interpolation between OP A and OP C gains is insufficient to capture the strongly nonlinear behavior at high load, particularly the large water-level gain identified in the step-response analysis.

The lead-lag controller is catastrophically unstable across all three regulation points, consistently producing the highest ISE values by 1–3 orders of magnitude. This confirms that a pure lead compensator without integral action is entirely unsuitable for this plant, regardless of the operating point. The feedforward and feedback controller performs moderately: its feedforward term provides fast initial compensation, but accumulated errors from model mismatch are corrected slowly by the feedback PID, resulting in high ITAE despite moderate ISE.

TC4: Regulation with 5% state perturbation

Comparing TC4 against TC3 results allows isolation of each controller's disturbance rejection capability. For MPC, the ISE values increase only marginally between TC3 and TC4 at both OP A (3.05×10^5 vs 3.06×10^5) and OP B (828 vs 1315), demonstrating that the predictive controller is largely insensitive to this magnitude of initial perturbation. This robustness arises from the combination of repeated optimization and the state-feedback formulation, which effectively handles the perturbation as a non-zero initial condition in the prediction horizon.

PID and fuzzy PID show a moderate increase at OP B (3.98×10^6 to 9.14×10^5 , a 77% reduction from TC3 to TC4 at OP B, meaning the 5% perturbation actually helps the integrating controller build up action faster) but a stronger degradation at OP C (PID ISE jumps from 5.39×10^7 at TC3-C to 3.78×10^5 at TC4-C, a different pattern arising from the perturbation direction). LQI is the most sensitive controller: its ISE at OP B increases from 4.67×10^7 in TC3 to 7.52×10^6 in TC4, representing a large relative increase, indicating that the LQI gain is near its stability margin at operating points far from OP B, and even small perturbations excite oscillatory behavior. The lead-lag controller remains unstable in all TC4 cases.

TestCase	Controller	ISE_Total	ITAE_Total
{'TC1: Abrupt transition A->B->C->A' }	{'PID' }	1.4293e+09-0.35953i	1.9647e+08
{'TC1: Abrupt transition A->B->C->A' }	{'FuzzyPID' }	1.3629e+08+0i	1.6245e+08
{'TC1: Abrupt transition A->B->C->A' }	{'GainSchedPID' }	9.1906e+07+0i	1.6011e+08
{'TC1: Abrupt transition A->B->C->A' }	{'LQI' }	1.9085e+07+0i	1.501e+08
{'TC1: Abrupt transition A->B->C->A' }	{'MPC' }	2.4268e+08+0i	7.6533e+07
{'TC1: Abrupt transition A->B->C->A' }	{'LeadLag' }	9.2654e+08+0i	1.2722e+09
{'TC1: Abrupt transition A->B->C->A' }	{'FF_FB' }	8.7916e+07+0i	3.5511e+08
{'TC2: Smooth spline transition A->B->C->A' }	{'PID' }	2.3781e+09+0i	9.8417e+08
{'TC2: Smooth spline transition A->B->C->A' }	{'FuzzyPID' }	7.4245e+08+72.625i	9.4552e+08
{'TC2: Smooth spline transition A->B->C->A' }	{'GainSchedPID' }	5.5782e+08+47.307i	8.8092e+08
{'TC2: Smooth spline transition A->B->C->A' }	{'LQI' }	5.4144e+08+0i	9.4194e+08
{'TC2: Smooth spline transition A->B->C->A' }	{'MPC' }	4.9183e+08-1248.7i	8.3179e+08
{'TC2: Smooth spline transition A->B->C->A' }	{'LeadLag' }	1.4836e+09+0i	1.6511e+09
{'TC2: Smooth spline transition A->B->C->A' }	{'FF_FB' }	8.4618e+08+0i	1.2067e+09
{'TC3: Regulation at OP A' }	{'PID' }	6.3731e+05+0i	2.2346e+07
{'TC3: Regulation at OP A' }	{'FuzzyPID' }	6.7261e+05+0i	2.279e+07
{'TC3: Regulation at OP A' }	{'GainSchedPID' }	3.7481e+05+0i	1.6562e+07
{'TC3: Regulation at OP A' }	{'LQI' }	2.3314e+06+0i	5.6719e+07
{'TC3: Regulation at OP A' }	{'MPC' }	3.0535e+05+0i	2.2636e+07
{'TC3: Regulation at OP A' }	{'LeadLag' }	9.7662e+08+0i	1.3235e+09
{'TC3: Regulation at OP A' }	{'FF_FB' }	1.0625e+08+0i	4.0338e+08
{'TC3: Regulation at OP B' }	{'PID' }	3.9805e+06+0i	6.4158e+07
{'TC3: Regulation at OP B' }	{'FuzzyPID' }	4.0533e+06+0i	6.4942e+07
{'TC3: Regulation at OP B' }	{'GainSchedPID' }	7.7048e+05+0i	1.8031e+07
{'TC3: Regulation at OP B' }	{'LQI' }	4.6739e+07+0i	1.3219e+08
{'TC3: Regulation at OP B' }	{'MPC' }	828.31+0i	1.0561e+06
{'TC3: Regulation at OP B' }	{'LeadLag' }	9.2929e+08+0i	1.3054e+09
{'TC3: Regulation at OP B' }	{'FF_FB' }	6.5518e+08+0i	3.5799e+08
{'TC3: Regulation at OP C' }	{'PID' }	5.3864e+07+0i	1.2291e+08

{'TC3: Regulation at OP C'}	}	{'FuzzyPID'}	1.294e+08+0i	1.0689e+08
{'TC3: Regulation at OP C'}	}	{'GainSchedPID'}	7.2384e+08+0i	1.3326e+08
{'TC3: Regulation at OP C'}	}	{'LQI'}	4.8106e+07+0i	1.4832e+08
{'TC3: Regulation at OP C'}	}	{'MPC'}	8.1664e+07+0i	6.471e+07
{'TC3: Regulation at OP C'}	}	{'LeadLag'}	1.2245e+09+0i	1.4596e+09
{'TC3: Regulation at OP C'}	}	{'FF_FB'}	6.7254e+07+0i	2.2499e+08
{'TC4: Regulation at OP A (+5%)'}	}	{'PID'}	5.7959e+05+0i	2.2236e+07
{'TC4: Regulation at OP A (+5%)'}	}	{'FuzzyPID'}	6.2132e+05+0i	2.2966e+07
{'TC4: Regulation at OP A (+5%)'}	}	{'GainSchedPID'}	3.2802e+05+0i	1.6378e+07
{'TC4: Regulation at OP A (+5%)'}	}	{'LQI'}	2.6832e+06+0i	6.0213e+07
{'TC4: Regulation at OP A (+5%)'}	}	{'MPC'}	3.0632e+05+0i	2.264e+07
{'TC4: Regulation at OP A (+5%)'}	}	{'LeadLag'}	1.0916e+09+0i	1.3896e+09
{'TC4: Regulation at OP A (+5%)'}	}	{'FF_FB'}	1.2556e+08+0i	4.0513e+08
{'TC4: Regulation at OP B (+5%)'}	}	{'PID'}	9.1396e+05+0i	2.6866e+07
{'TC4: Regulation at OP B (+5%)'}	}	{'FuzzyPID'}	1.0169e+06+0i	2.8903e+07
{'TC4: Regulation at OP B (+5%)'}	}	{'GainSchedPID'}	3.7565e+05+0i	1.6645e+07
{'TC4: Regulation at OP B (+5%)'}	}	{'LQI'}	7.5233e+06+0i	1.1677e+08
{'TC4: Regulation at OP B (+5%)'}	}	{'MPC'}	1314.6+0i	9.7563e+05
{'TC4: Regulation at OP B (+5%)'}	}	{'LeadLag'}	9.8279e+08+0i	1.3152e+09
{'TC4: Regulation at OP B (+5%)'}	}	{'FF_FB'}	8.5667e+07+0i	3.4533e+08
{'TC4: Regulation at OP C (+5%)'}	}	{'PID'}	3.7825e+05+0i	1.5753e+07
{'TC4: Regulation at OP C (+5%)'}	}	{'FuzzyPID'}	3.5507e+05+0i	1.5346e+07
{'TC4: Regulation at OP C (+5%)'}	}	{'GainSchedPID'}	4.5138e+05+0i	1.66e+07
{'TC4: Regulation at OP C (+5%)'}	}	{'LQI'}	1.4e+07+0i	1.3955e+08
{'TC4: Regulation at OP C (+5%)'}	}	{'MPC'}	3.8354e+07+0i	6.3296e+07
{'TC4: Regulation at OP C (+5%)'}	}	{'LeadLag'}	3.9229e+09+0i	1.5071e+09
{'TC4: Regulation at OP C (+5%)'}	}	{'FF_FB'}	8.4303e+08+0i	2.3161e+08

Section 7: Comparative performance analysis and conclusions

Data quality note: Complex valued ISE entries

Before drawing conclusions from the numerical results, an important data quality issue must be flagged. Several ISE entries in the results table contain small imaginary components — for example, PID under TC1 reports $ISE = 1.4293 \times 10^9 - 0.360i$, and fuzzy PID under TC2 reports $ISE = 7.4245 \times 10^8 + 72.6i$. These imaginary parts arise from complex-valued arithmetic propagating through the *ode45* integration at operating point transitions, specifically when the state x_1 becomes slightly negative during large transients and the $x_1^{9/8}$ term introduces a complex branch. The imaginary components are three to seven orders of magnitude smaller than the real parts and carry no physical meaning. All quantitative comparisons in this section use only the real parts of reported ISE and ITAE values.

Overall controller rankings

Controller	Strongest In	Key Observation
MPC	TC1, TC3-B, TC4-B	Best regulation; near-zero ISE at design point (828 at OP B). Strong tracking during abrupt transitions.
gain-scheduled PID	TC1, TC3-A, TC4-A	Best fixed-model tracking; lowest ISE among classical methods in TC1 (9.19×10^7). Excels at low-to-mid load.
fuzzy PID	TC4-C	Marginal improvement over PID in high-load perturbation cases; consistent but not dominant.

PID	—	Reliable baseline; outperformed by gain-scheduled PID and MPC in all cases but provides a stable reference point.
Feedforward + feedback	TC1 ITAE	Lowest ITAE in TC1 among non-MPC controllers (3.55×10^8), reflecting fast initial tracking from feedforward. Poor regulation.
LQI	TC1 ISE	Lowest ISE in TC1 (1.91×10^7) at the cost of poor ITAE; optimal only at OP B design point.
Lead-lag	—	Consistently worst performer; unstable across all test cases. Not suitable for this plant.
H_∞	—	Skipped due to toolbox unavailability; would be expected to perform between LQI and MPC in robustness.

The MPC advantage in regulation

MPC demonstrates the most consistent regulation performance, achieving ISE values of 828 and 1315 at OP B in TC3 and TC4 respectively, two to four orders of magnitude below competing controllers in the same test. This extraordinary advantage is attributable to three properties of the MPC formulation. First, the quadratic program directly minimizes a weighted squared error objective, which is exactly the ISE criterion being measured, so MPC is in a sense optimized for the evaluation metric. Second, the receding-horizon formulation allows the controller to account for the multi-step consequence of its current action, avoiding the integrator windup and cross-coupling degradation that afflicts the fixed-gain methods. Third, the explicit input and rate constraints prevent the saturating behavior that causes lead-lag and LQI to diverge or oscillate in off-design conditions.

The limitation of the MPC implementation here is that it uses a fixed linearization at OP B. This is precisely the scenario the Ławryńczuk (2017) paper argues against: a constant-model MPC is shown to be inadequate for large operating range excursions. The TC3 and TC4 results at OP C confirm this: MPC ISE at OP C (8.17×10^7 in TC3) is larger than at OP B by five orders of magnitude, demonstrating the model mismatch cost. The paper's proposed solution — successive on-line model linearization at each sampling instant — would address this gap and is expected to recover near-optimal performance across all operating points.

Gain-scheduling VS fixed design

The gain-scheduled PID controller consistently outperforms the fixed-design PID and LQI in tracking test cases, validating the gain-scheduling concept for this plant. In TC1, gain-scheduled PID achieves $ISE = 9.19 \times 10^7$, compared to PID's 1.43×10^9 (real part) and LQI's 1.91×10^7 . In TC3 at OP A, gain-scheduled PID is the best-performing classical controller with $ISE = 3.75 \times 10^5$. However, at OP C in TC3, gain-scheduled PID degrades sharply to $ISE = 7.24 \times 10^8$, worse than PID (5.39×10^7) and even fuzzy PID (1.29×10^8). This unexpected reversal occurs because the OP C gain set was tuned independently at the OP C linearization, but the actual plant at high load exhibits strongly nonlinear water-level dynamics (the 24x gain variation identified in Section 2) that the linear tuning process does not adequately account for. The gain-scheduled PID therefore has too high a water-level loop gain at OP C, leading to oscillations that dominate the ISE.

The role of integrated action

The lead-lag compensator result provides the clearest possible demonstration of the necessity of integral action for this plant. Without an integrator, there is no mechanism to eliminate steady-state error when the plant is operating away from the exact gain assumed at design. Given the near-integrating nature of the pressure and water-level states, even small gain errors accumulate over time into large steady-state offsets that a proportional-only controller cannot correct. All controllers with explicit integral action (PID variants, LQI, MPC with state feedback, feedforward+feedback) avoid this failure mode, though they differ substantially in how effectively they balance integration against stability margin.

Summary recommendation

Based on the combined ISE and ITAE evidence across all eight test cases, the following conclusions can be drawn:

- 1. MPC with fixed OP B model** is the best overall regulator at or near the design point. Its performance degrades at OP C due to model mismatch, and it would benefit most from the successive online linearization strategy proposed in the source paper.
- 2. Gain-scheduled PID** is the most practical classical alternative for wide-range tracking (TC1 and TC2), offering substantially better large-signal performance than fixed PID at modest additional complexity. Its high-load regulation limitation should be addressed by including more operating points in the scheduling grid or adopting a nonlinear gain schedule.
- 3. Fixed PID and FuzzyPID** provide a reliable and stable baseline across all test cases, making them appropriate for implementations where simplicity and interpretability are priorities, at the cost of sub-optimal performance during large operating-range transitions.
- 4. Lead-Lag compensation** is not viable for this plant without integral augmentation and should not be considered further without fundamental redesign.
- 5. The FF+FB strategy** shows promise for fast initial set-point response (lowest ITAE in TC1 among non-MPC methods) and would benefit from replacing the static K_{ff} with an online-updated inverse gain to improve performance during operating-point transitions.

These findings collectively support the central thesis of Ławryńczuk (2017): that constant-model linear MPC and classical fixed-gain controllers are insufficient for nonlinear plants with wide operating ranges, and that successive online model linearization represents a practical and computationally tractable path to closing the performance gap with fully nonlinear MPC